Chapter 10. Discrimination and Classification

Bayes rule, LDA/QDA, Fisher's discriminant, logistic regression, assessing classifiers

Contents

1 The Bayes Classification Rule **2**

1.1 Setting 2

2 Normal Populations: LDA and QDA **2**

2.1 Common covariance: linear discriminant analysis 2

2.2 Different covariances: quadratic discriminant analysis 3

3 Fisher's Linear Discriminant **3**

4 Logistic Regression **4**

5 Simple Classifiers in High Dimensions **4**

6 Assessment of Classifiers **5**

7 Selected exercises **5**

1 The Bayes Classification Rule

Motivation. Classification is a decision problem: assign a new observation to a group so as to minimize expected misclassification cost. The Bayes rule compares prior-weighted densities and is optimal; every practical classifier estimates it.

1.1 Setting

Let $Y \in \{1, \dots, K\}$ be the group and $\mathbf{X} \in \mathbb{R}^p$ the features, with priors $P(Y = k) = \pi_k$ and $\mathbf{X} \mid Y = k \sim f_k$. For $K = 2$: $\mathbf{X} \mid Y = 1 \sim f_1(\mathbf{x})$, $\mathbf{X} \mid Y = 2 \sim f_2(\mathbf{x})$, $P(Y = 1) = \pi_1$, and

$$\eta(\mathbf{x}) := P(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \frac{f_1(\mathbf{x})\pi_1}{f_1(\mathbf{x})\pi_1 + f_2(\mathbf{x})\pi_2}.$$

The misclassification probability of a classifier δ (or ϕ) : $\mathbb{R}^p \to \{1, \dots, K\}$ is $P(\phi(\mathbf{X}) \neq Y) = \sum_k P(\phi(\mathbf{X}) \neq k \mid Y = k)P(Y = k)$.

Theorem 10.1: optimality of the Bayes rule

$\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x}) = \arg\max_k \pi_k f_k(\mathbf{x})$ minimizes the misclassification probability: $P(\phi_{\text{Bayes}}(\mathbf{X}) \neq Y) \leq P(\phi(\mathbf{X}) \neq Y)$ for every ϕ.

$K = 2$ **[Supplement].**

$$\begin{aligned}
\text{Risk}(\delta) &= \mathrm{E}\big[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}}\big] = \mathrm{E}_Y\Big[\mathrm{E}_{\mathbf{X}|Y}\big[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}} \mid Y\big]\Big] \\
&= \pi_1 \mathrm{E}_{\mathbf{X}|Y=1}\big[\mathbf{1}_{\{\delta(\mathbf{X})\neq 1\}} \mid Y = 1\big] + \pi_2 \mathrm{E}_{\mathbf{X}|Y=2}\big[\mathbf{1}_{\{\delta(\mathbf{X})\neq 2\}} \mid Y = 2\big] \\
&= \int \underbrace{\Big[\pi_1 \mathbf{1}_{\{\delta(\mathbf{x})\neq 1\}} f_1(\mathbf{x}) + \pi_2 \mathbf{1}_{\{\delta(\mathbf{x})\neq 2\}} f_2(\mathbf{x})\Big]}_{=:h(\mathbf{x})} d\mathbf{x}.
\end{aligned}$$

At each $\mathbf{x}$, $\delta(\mathbf{x}) = 1$ gives $h(\mathbf{x}) = \pi_2 f_2(\mathbf{x})$ and $\delta(\mathbf{x}) = 2$ gives $h(\mathbf{x}) = \pi_1 f_1(\mathbf{x})$. To minimize the integral choose the smaller pointwise: if $\pi_2 f_2(\mathbf{x}) < \pi_1 f_1(\mathbf{x})$ choose $\delta(\mathbf{x}) = 1$; if $>$, choose $\delta(\mathbf{x}) = 2$. Thus $\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x})$. For general K the same argument minimizes $1 - P(Y = \phi(\mathbf{x}) \mid \mathbf{x})$ pointwise. ■

Remark 10.2: misclassification costs

If $c(2 \mid 1)$ is the cost of classifying a member of group 1 into group 2, the rule minimizing expected cost classifies into group 1 when $\dfrac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \dfrac{c(1 \mid 2)}{c(2 \mid 1)} \dfrac{\pi_2}{\pi_1}$.

2 Normal Populations: LDA and QDA

2.1 Common covariance: linear discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ ($\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}$), then $\log\{\pi_k f_k(\mathbf{x})\} = -\frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \log \pi_k +$ const, and the first term does not depend on k, so

$$\phi(\mathbf{x}) = \arg\min_k \delta_k(\mathbf{x}), \qquad \delta_k(\mathbf{x}) = b_{0k} - \mathbf{b}_k^\top \mathbf{x}, \quad b_{0k} = \tfrac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k - \log(\pi_k), \quad \mathbf{b}_k = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k.$$

The discriminant function is linear in $\mathbf{x}$: **linear discriminant analysis (LDA)**. For $K = 2$, $\delta_1 < \delta_2$ rearranges to

$$\phi(\mathbf{x}) = 1 \;\text{ if }\; \mathbf{b}^\top\Big(\mathbf{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2}{2}\Big) < \log\Big(\frac{\pi_1}{\pi_2}\Big), \qquad \mathbf{b} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1).$$

$K = 2$ **[Supplement].** $\delta_1 < \delta_2 \iff (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2) > \log(\pi_2/\pi_1)$. Since $\boldsymbol{\mu}_1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$, this is $-\mathbf{b}^\top(\mathbf{x} - \frac{\boldsymbol{\mu}_1+\boldsymbol{\mu}_2}{2}) > \log(\pi_2/\pi_1)$. ■

The sample version plugs in $\boldsymbol{\mu}_k \to \bar{\mathbf{x}}_k$, $\boldsymbol{\Sigma} \to \mathbf{S}_p$ (pooled covariance), $\pi_k \to n_k/n$: $\phi(\mathbf{x}) = \arg\max_k(b_{0k} - \mathbf{b}_k^\top\mathbf{x})$-type rule with $\hat{\mathbf{b}}_k = \mathbf{S}_p^{-1}\bar{\mathbf{x}}_k$, and for $K = 2$, $\hat{\mathbf{b}} = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$.

2.2 Different covariances: quadratic discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$,

$$\phi(\mathbf{x}) = \arg\max_k\Big\{-\tfrac{1}{2}\log|\boldsymbol{\Sigma}_k| - \tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top\boldsymbol{\Sigma}_k^{-1}(\mathbf{x} - \boldsymbol{\mu}_k) + \log\pi_k\Big\} = \arg\max_k\,(b_{0k} - \mathbf{b}_k^\top\mathbf{x} + \mathbf{x}^\top\mathbf{C}_k\mathbf{x})$$

with $\mathbf{C}_k = -\frac{1}{2}\boldsymbol{\Sigma}_k^{-1}$. The boundaries are quadratic surfaces: **QDA**. It needs $K \cdot p(p+1)/2$ more parameters, so with small samples LDA is often more stable (bias–variance trade-off).

3 Fisher's Linear Discriminant

Motivation. Fisher's approach needs no distributional assumption: find the projection that best separates the groups relative to the within-group spread. Under normality with equal covariances it coincides with the Bayes rule.

Without distributional assumptions, find the projection that best separates the groups.

Definition 10.3: Fisher's LDA direction ($K = 2$)

$$\mathbf{u}_0 = \arg\max_{\mathbf{u}\in\mathbb{R}^p} \frac{(\mathbf{u}^\top\bar{\mathbf{x}}_1 - \mathbf{u}^\top\bar{\mathbf{x}}_2)^2}{\mathbf{u}^\top\mathbf{S}_p\mathbf{u}}:$$

squared difference of projected means over projected within-group variance.

Theorem 10.4: solution

$\mathbf{u}_0 \propto \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$ (equivalently $\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$), and the maximum is $\mathbf{d}^\top\mathbf{S}_p^{-1}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, the squared sample Mahalanobis distance.

Proof [Supplement]. In general, to maximize $\frac{\mathbf{u}^\top\mathbf{A}\mathbf{u}}{\mathbf{u}^\top\mathbf{B}\mathbf{u}}$ ($\mathbf{B} \succ 0$) use $L(\mathbf{u}, \lambda) = \mathbf{u}^\top\mathbf{A}\mathbf{u} - \lambda(\mathbf{u}^\top\mathbf{B}\mathbf{u} - 1)$: $\partial L/\partial\mathbf{u} = \mathbf{0} \Rightarrow \mathbf{A}\mathbf{u} = \lambda\mathbf{B}\mathbf{u} \Rightarrow \mathbf{B}^{-1}\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$, so the top eigenvector of $\mathbf{B}^{-1}\mathbf{A}$ is optimal. With $\mathbf{A} = \mathbf{d}\mathbf{d}^\top$ (rank 1) and $\mathbf{B} = \mathbf{S}_p$, $\mathbf{S}_p^{-1}\mathbf{d}\mathbf{d}^\top$ has a single nonzero eigenvalue $\mathbf{d}^\top\mathbf{S}_p^{-1}\mathbf{d}$ with unique eigenvector $\mathbf{S}_p^{-1}\mathbf{d}$ (rank 1). Directly: with $\mathbf{v} = \mathbf{S}_p^{1/2}\mathbf{u}$, Cauchy–Schwarz gives $(\mathbf{u}^\top\mathbf{d})^2/\mathbf{u}^\top\mathbf{S}_p\mathbf{u} = (\mathbf{v}^\top\mathbf{S}_p^{-1/2}\mathbf{d})^2/\mathbf{v}^\top\mathbf{v} \le \mathbf{d}^\top\mathbf{S}_p^{-1}\mathbf{d}$. ■

Remark 10.5: Fisher's rule = LDA with equal priors

With the discriminant score $y_i = \mathbf{b}_Y^\top\mathbf{x}_i$, $\mathbf{b}_Y = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$, and the cut-off $b_{0Y} = \mathbf{b}_Y^\top(\bar{\mathbf{x}}_1 +$

$\bar{\mathbf{x}}_2)/2$,

$$\text{“}\mathbf{b}_Y^\top \mathbf{y} - b_{0Y} < 0 \Rightarrow 1\text{”} \iff \text{“}\big[\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)\big]^\top \Big(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}\Big) < 0 \Rightarrow 1\text{”},$$

which is sample LDA with $\pi_1 = \pi_2$. A rule derived without normality coincides with the Bayes rule under normality.

Several groups ($K \geq 3$): $\max_{\mathbf{u}} \dfrac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{W}\mathbf{u}}$ ($\mathbf{B}, \mathbf{W}$ the MANOVA between/within SSP matrices) is solved by the eigenvectors of $\mathbf{W}^{-1}\mathbf{B}$; there are at most $\min(p, K-1)$ nonzero eigenvalues, the same η_i that appear in the Wilks/Pillai statistics of Ch. 6.

4 Logistic Regression

For $K = 2$, $Y \in \{0, 1\}$, model the posterior directly:

$$\operatorname{logit}(\eta(\mathbf{x})) := \log\Big(\frac{\eta(\mathbf{x})}{1 - \eta(\mathbf{x})}\Big) = b + \boldsymbol{\beta}^\top \mathbf{x}, \qquad \eta(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x}).$$

Classification: “$\eta(\mathbf{x}) > \frac{1}{2} \Rightarrow 1$” $\iff$ “$f(\mathbf{x}) = b + \boldsymbol{\beta}^\top \mathbf{x} > 0 \Rightarrow 1$” (a linear boundary). The log-likelihood is

$$\begin{aligned} \ell(b, \boldsymbol{\beta}) &= \sum_{i=1}^{n} \big[y_i \log(\eta(\mathbf{x}_i; b, \boldsymbol{\beta})) + (1 - y_i) \log\{1 - \eta(\mathbf{x}_i; b, \boldsymbol{\beta})\}\big] \\ &= \sum_{i=1}^{n} \big[y_i (b + \boldsymbol{\beta}^\top \mathbf{x}_i) - \log\{1 + \exp(b + \boldsymbol{\beta}^\top \mathbf{x}_i)\}\big], \end{aligned}$$

which is concave and maximized by Newton–Raphson (IRLS). In the equal-covariance normal model $\log \frac{\pi_1 f_1}{\pi_0 f_0}$ is exactly linear in $\mathbf{x}$, so the logistic model holds; LDA models the joint distribution, logistic regression only the conditional one.

Derivation: the second equality and the derivatives

$\log \eta = (b + \boldsymbol{\beta}^\top \mathbf{x}) - \log(1 + e^{b + \boldsymbol{\beta}^\top \mathbf{x}})$ and $\log(1 - \eta) = -\log(1 + e^{b + \boldsymbol{\beta}^\top \mathbf{x}})$, so $y \log \eta + (1 - y) \log(1 - \eta) = y(b + \boldsymbol{\beta}^\top \mathbf{x}) - \log(1 + e^{b + \boldsymbol{\beta}^\top \mathbf{x}})$. The gradient is $\sum_i (y_i - \eta_i)(1, \mathbf{x}_i^\top)^\top$ and the Hessian $-\sum_i \eta_i(1 - \eta_i)(1, \mathbf{x}_i^\top)^\top (1, \mathbf{x}_i^\top) \preceq 0$.

5 Simple Classifiers in High Dimensions

If $p > n$, $\mathbf{S}_p$ is singular and LDA cannot be used (e.g. the Golub data).

- **Nearest centroid**: LDA with $\boldsymbol{\Sigma} = \mathbf{I}$. $\phi(\mathbf{x}) = 1$ if $\mathbf{b}^\top(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}) < 0$, $\mathbf{b} = \bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1$ (assign to the closer mean).
- **Naive Bayes classifier (diagonal LDA)**: assume independent variables and replace $\mathbf{S}_p$ by $\mathbf{D}_p = \operatorname{diag}(\mathbf{S}_p)$: $\phi(\mathbf{x}) = 1$ if $\mathbf{b}^\top(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}) < 0$, $\mathbf{b} = \mathbf{D}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$.
- **k-nearest neighbors**: majority vote among the k training points closest to $\mathbf{x}$; non-parametric, high variance for small k and high bias for large k.

6 Assessment of Classifiers

- True error rate $P(\phi(\mathbf{X}) \neq Y) = \sum_{k=1}^{K} P(\phi(\mathbf{X}) \neq k \mid Y = k)P(Y = k)$; ϕ is optimal if $P(\phi(\mathbf{X}) \neq Y) \leq P(\varphi(\mathbf{X}) \neq Y)$ for all φ.
- Empirical misclassification rate $\hat{P}(\hat{\phi}_D(\mathbf{X}) \neq Y \mid D) = \frac{1}{M} \sum_{i=1}^{M} \mathbf{1}_{\{\hat{\phi}(\mathbf{x}_i) \neq y_i\}}$.
- **Training and test sets**: build $\hat{\phi}$ from $\mathbf{X}_{\text{tr}}$ and estimate the misclassification rate on $\mathbf{X}_{\text{test}}$. The training error (APER) is optimistically biased.
- **Confusion matrix**: actual group by predicted group (training and testing confusion matrices).
- **K-fold cross-validation**: split into K parts, use each in turn as the test set and average. $K \uparrow \Rightarrow$ bias $\downarrow$, variance $\uparrow$; $K \downarrow \Rightarrow$ the reverse. $K = n$ is LOOCV (Lachenbruch's holdout).

7 Selected exercises

Motivation. The exercises prove the cost-weighted Bayes rule stated above, work out classification regions for non-normal densities, show how the total separation of several groups splits over Fisher's discriminants, and explain when logistic regression has no maximum likelihood estimate.

Exercise 10.1 (The minimum expected cost rule; J&W 11.3, 11.4)

For two populations with densities f_1, f_2, priors π_1, π_2 and costs $c(2 \mid 1), c(1 \mid 2) \geq 0$, a rule is a partition $\mathbb{R}^p = R_1 \cup R_2$ and

$$\mathrm{ECM}(R_1) = c(2 \mid 1)\pi_1 \int_{R_2} f_1(\mathbf{x})\, d\mathbf{x} + c(1 \mid 2)\pi_2 \int_{R_1} f_2(\mathbf{x})\, d\mathbf{x}.$$

(a) Show that ECM is minimized by $R_1^* = \{\mathbf{x} : c(2 \mid 1)\pi_1 f_1(\mathbf{x}) \geq c(1 \mid 2)\pi_2 f_2(\mathbf{x})\}$.
(b) With $c(2 \mid 1) = 50$, $c(1 \mid 2) = 100$, $\pi_2 = 0.2$, classify an item with $f_1(\mathbf{x}) = 0.3$ and $f_2(\mathbf{x}) = 0.5$.

Solution.

(a)

Step 1. Since $\int_{R_2} f_1 = 1 - \int_{R_1} f_1$,

$$\mathrm{ECM}(R_1) = c(2 \mid 1)\pi_1 + \int_{R_1} h(\mathbf{x})\, d\mathbf{x}, \qquad h(\mathbf{x}) = c(1 \mid 2)\pi_2 f_2(\mathbf{x}) - c(2 \mid 1)\pi_1 f_1(\mathbf{x}).$$

Step 2. For any measurable R_1, splitting R_1 according to the sign of h,

$$\int_{R_1} h = \int_{R_1 \cap \{h \leq 0\}} h + \int_{R_1 \cap \{h > 0\}} h \;\geq\; \int_{R_1 \cap \{h \leq 0\}} h \;\geq\; \int_{\{h \leq 0\}} h = \int_{R_1^*} h,$$

because adding the points of $\{h \leq 0\}$ outside R_1 can only decrease the integral. Hence $\mathrm{ECM}(R_1) \geq \mathrm{ECM}(R_1^*)$. Where $f_2 > 0$ this is the ratio form $f_1/f_2 \geq \{c(1 \mid 2)/c(2 \mid 1)\}(\pi_2/\pi_1)$; points with $h = 0$ may be assigned either way.

(b) The threshold is $(100/50)(0.2/0.8) = 0.5$ and $f_1/f_2 = 0.6 \geq 0.5$, so the item is assigned to population 1, although $f_2 > f_1$: the larger prior of population 1 outweighs the doubled cost. ■

Exercise 10.2 (Triangular densities; J&W 11.7)

Let $f_1(x) = 1-|x|$ for $|x| \le 1$ and $f_2(x) = 1-|x-0.5|$ for $-0.5 \le x \le 1.5$ (both 0 elsewhere), with equal costs.
(a) Find the classification regions for $\pi_1 = \pi_2$ and the total probability of misclassification.
(b) Find the regions for $\pi_1 = 0.2$.

Solution. By Exercise 10.1 with equal costs, assign x to population 1 iff $\pi_1 f_1(x) \ge \pi_2 f_2(x)$. On $[-1,-0.5)$ only f_1 is positive (population 1) and on $(1,1.5]$ only f_2 (population 2); it remains to treat $[-0.5,1]$, where $f_2(x) = 0.5+x$ for $x \le 0.5$ and $f_2(x) = 1.5-x$ for $x \ge 0.5$.

(a)

Step 1. On $[-0.5,1]$, $f_1 \ge f_2 \iff |x| \le |x-0.5| \iff x \le 0.25$. So $R_1 = [-1, 0.25]$ and $R_2 = (0.25, 1.5]$.

Step 2. $P(2 \mid 1) = \int_{0.25}^{1}(1-x)\,dx = 0.75^2/2 = 0.28125$ and $P(1 \mid 2) = \int_{-0.5}^{0.25}(0.5+x)\,dx = 0.75^2/2 = 0.28125$, so the total probability of misclassification is $\frac{1}{2}(0.28125 + 0.28125) = 0.28125$.

(b) Now population 1 needs $0.2 f_1 \ge 0.8 f_2$, i.e. $f_1 \ge 4 f_2$.

Step 1. On $[-0.5, 0]$: $1 + x \ge 4(0.5 + x) \iff x \le -1/3$.

Step 2. On $[0, 0.5]$: $1 - x \ge 2 + 4x \iff x \le -0.2$, impossible. On $[0.5, 1]$: $1 - x \ge 6 - 4x \iff x \ge 5/3$, impossible.

Step 3. Hence $R_1 = [-1, -1/3]$ and $R_2 = (-1/3, 1.5]$: the smaller prior shifts the boundary from 0.25 to $-1/3$. ■

Exercise 10.3 (Total separation and Fisher's discriminants; J&W 11.22)

For g groups with means $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_g$, common covariance $\boldsymbol{\Sigma} \succ 0$ and $\bar{\boldsymbol{\mu}} = \frac{1}{g}\sum_i \boldsymbol{\mu}_i$, let $\mathbf{B}_\mu = \sum_i (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})^\top$ and define the total separation

$$\Delta_S^2 = \sum_{i=1}^{g} (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}}).$$

Let $\lambda_1 \ge \cdots \ge \lambda_s > 0$ be the nonzero eigenvalues of $\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu$ and $Y_k = \mathbf{e}_k^\top \mathbf{X}$ the discriminants, with $\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu \mathbf{e}_k = \lambda_k \mathbf{e}_k$ and $\mathbf{e}_k^\top \boldsymbol{\Sigma} \mathbf{e}_l = \delta_{kl}$. Show $\Delta_S^2 = \lambda_1 + \cdots + \lambda_s$, and that the first r discriminants carry separation $\sum_{k \le r} \sum_i (\mathrm{E}_i Y_k - \bar{\mu}_{Y_k})^2 = \lambda_1 + \cdots + \lambda_r$.

Solution.

Step 1. $\Delta_S^2 = \sum_i \mathrm{tr}\{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})^\top\} = \mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu) = \mathrm{tr}(\mathbf{A})$, where $\mathbf{A} = \boldsymbol{\Sigma}^{-1/2}\mathbf{B}_\mu\boldsymbol{\Sigma}^{-1/2}$ is symmetric and positive semidefinite.

Step 2. $\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu = \boldsymbol{\Sigma}^{-1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2}$ is similar to $\mathbf{A}$, so they have the same eigenvalues, and $\mathrm{tr}\mathbf{A}$ is the sum of the eigenvalues; the zero ones contribute nothing, so $\Delta_S^2 = \lambda_1 + \cdots + \lambda_s$.

Step 3. Let $\mathbf{A} = \sum_k \lambda_k \mathbf{f}_k \mathbf{f}_k^\top$ with orthonormal $\mathbf{f}_k$ and $\mathbf{e}_k = \boldsymbol{\Sigma}^{-1/2}\mathbf{f}_k$. Then $\boldsymbol{\Sigma}^{-1}\mathbf{B}_\mu\mathbf{e}_k = \boldsymbol{\Sigma}^{-1/2}\mathbf{A}\mathbf{f}_k = \lambda_k \mathbf{e}_k$ and $\mathbf{e}_k^\top\boldsymbol{\Sigma}\mathbf{e}_l = \mathbf{f}_k^\top\mathbf{f}_l = \delta_{kl}$, so these are the discriminants: $\mathrm{Var}(Y_k) = 1$ and $\mathrm{Cov}(Y_k, Y_l) = 0$.

Step 4. The group means of Y_k are $\mathbf{e}_k^\top\boldsymbol{\mu}_i$ with average $\mathbf{e}_k^\top\bar{\boldsymbol{\mu}}$, so

$$\sum_{i=1}^{g} \{\mathbf{e}_k^\top(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}})\}^2 = \mathbf{e}_k^\top\mathbf{B}_\mu\mathbf{e}_k = \mathbf{f}_k^\top\mathbf{A}\mathbf{f}_k = \lambda_k.$$

Summing over $k \le r$ gives $\lambda_1 + \cdots + \lambda_r$; for $r = s$ this is all of Δ_S^2. Thus $(\lambda_1 + \cdots + \lambda_r)/\Delta_S^2$ is the share of separation kept by the first r discriminants. ■

Exercise 10.4 (Logistic regression: linearity and separation)

(a) If $\mathbf{X} \mid Y = k \sim N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$, $k = 0, 1$, show $\operatorname{logit} \eta(\mathbf{x}) = b + \boldsymbol{\beta}^\top \mathbf{x}$ with $\boldsymbol{\beta} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$ and $b = \log(\pi_1/\pi_0) - \frac{1}{2}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$.
(b) Suppose the data are completely separated: there is $(b_0, \boldsymbol{\beta}_0)$ with $z_i = b_0 + \boldsymbol{\beta}_0^\top \mathbf{x}_i > 0$ when $y_i = 1$ and $z_i < 0$ when $y_i = 0$. Show that the log-likelihood $\ell(b, \boldsymbol{\beta})$ has no maximizer.

Solution [Supplement].

(a) $\operatorname{logit} \eta(\mathbf{x}) = \log \dfrac{\pi_1 f_1(\mathbf{x})}{\pi_0 f_0(\mathbf{x})} = \log \dfrac{\pi_1}{\pi_0} - \dfrac{1}{2}\{(\mathbf{x}-\boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_1) - (\mathbf{x}-\boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_0)\}$. The terms $\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x}$ cancel, leaving $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_0)$, and $\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_0 = (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$ by symmetry of $\boldsymbol{\Sigma}^{-1}$.

(b)

Step 1 ($\ell < 0$). Every term $y_i \log \eta_i + (1 - y_i) \log(1 - \eta_i)$ is negative because $0 < \eta_i < 1$, so $\ell(b, \boldsymbol{\beta}) < 0$ for every parameter.

Step 2 ($\sup \ell = 0$). Along $(tb_0, t\boldsymbol{\beta}_0)$, $t \to \infty$: if $y_i = 1$, $\eta_i = 1/(1 + e^{-tz_i}) \to 1$ since $z_i > 0$; if $y_i = 0$, $1 - \eta_i = 1/(1 + e^{tz_i}) \to 1$ since $z_i < 0$. Every term tends to 0, so $\ell(tb_0, t\boldsymbol{\beta}_0) \to 0$.

Step 3. The supremum 0 is approached but, by Step 1, never attained, so no MLE exists. Newton–Raphson then drives $\|\boldsymbol{\beta}\|$ to infinity; in practice one uses a penalty (e.g. ridge) or reports the separation. LDA, which models $\mathbf{X}$ given Y, is unaffected. ■